



How Anxiety/Depression levels differed between states and regions following COVID-19

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Background

“The spread of disease and increase in deaths during large outbreaks of transmissible diseases is often associated with fear and grief.” - CDC

Data Characteristics

- Anxiety and Depression measured as a percentage of population exhibiting symptoms
- Data collected every 2 weeks from 4/23/2020 to 9/20/2023
- Regions sorted by US Census Definition



Patient Health Questionnaire (PHQ-4)

During the past 7 days, respondents had been bothered by:



Inclusion

Adults with symptoms occurring more than one half of the days or nearly every day were classified as having symptoms



Question 1

Feeling nervous, anxious, or on edge



Question 2

Not being able to stop or control worrying



Question 3

Having little interest or pleasure in doing things



Question 4

Feeling down, depressed, or hopeless.



Research Question

How did anxiety and depression levels differ between states and regions following the outbreak of COVID-19 in the United States?

01

Cultural Differences

Different parts of the US have different cultural norms/customs

02

Response to COVID

Some states, such as South Dakota, had less restrictive lockdowns or very little at all

03

Differences in Rates

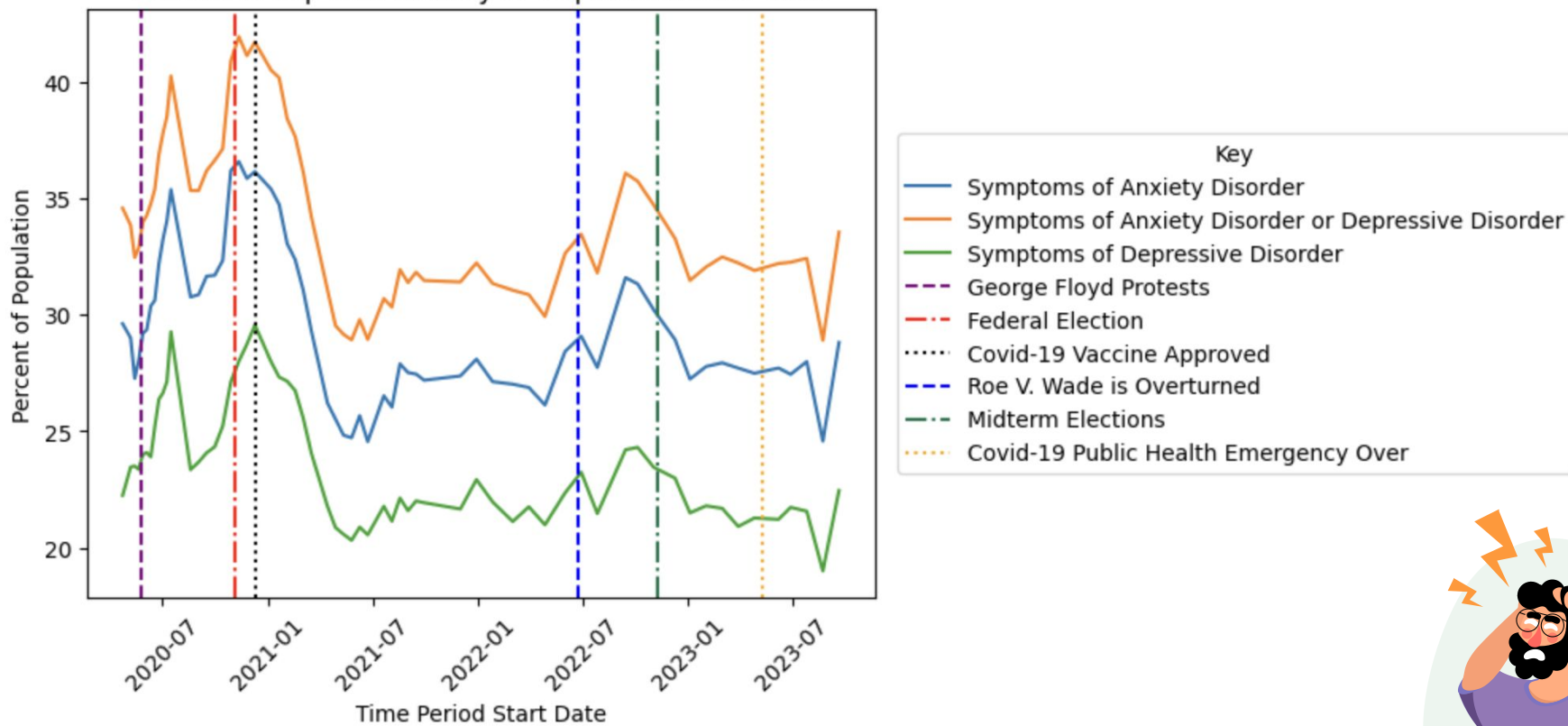
Some states had much higher per capita death rates
TX (2020): 105.2/100k
WA (2020): 36.7/100k

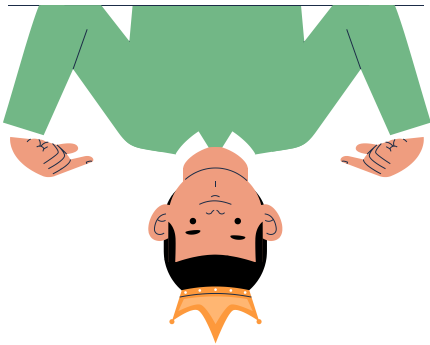
Variables Used

	Indicator	Whether each observation was measuring anxiety, depression, or either. A “Symptoms of” variable was also created to describe this more succinctly.
	State/ID Code/Region	Which state was being measured. An additional variable was created to store the state ID code (e.g. CA for California), and another was created to store the region of the US based on the census.
	Time Period + Start/End Date	The time period variable measures observations in order (starting with 4/23/2020). The start and end date variables measure the specific start and end of each time period.
	Percent of Population	What percent of the population of each state was experiencing symptoms of the given indicator.

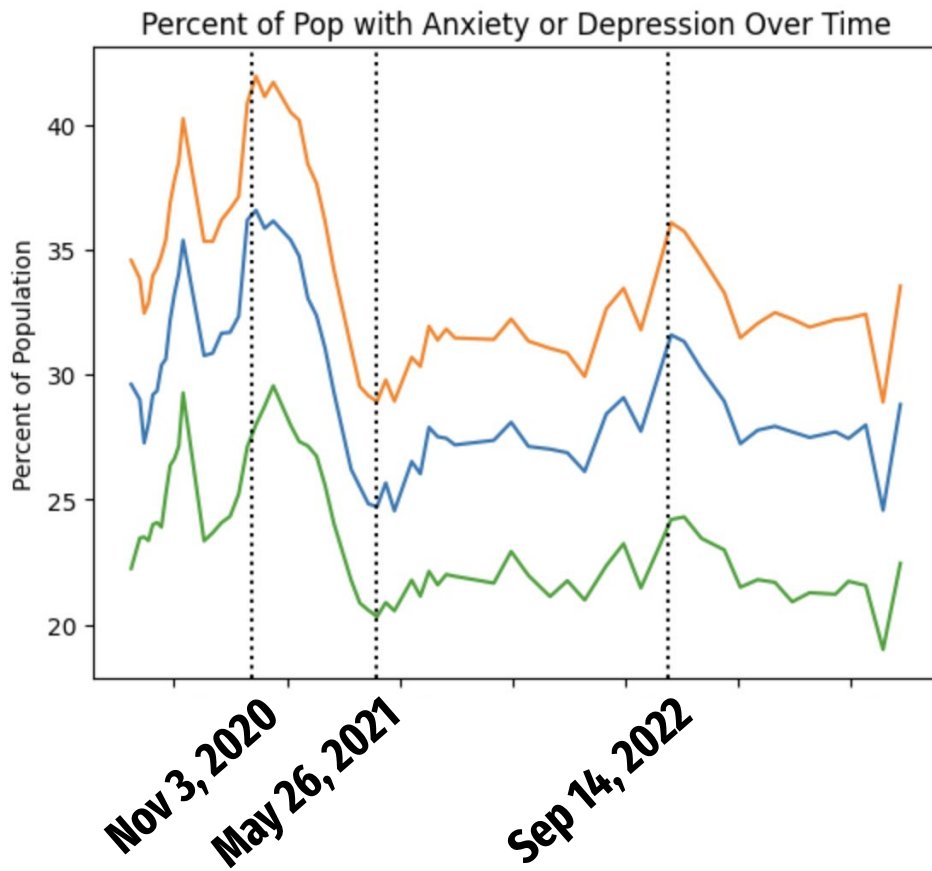
Results

Percent of Pop with Anxiety or Depression Over Time

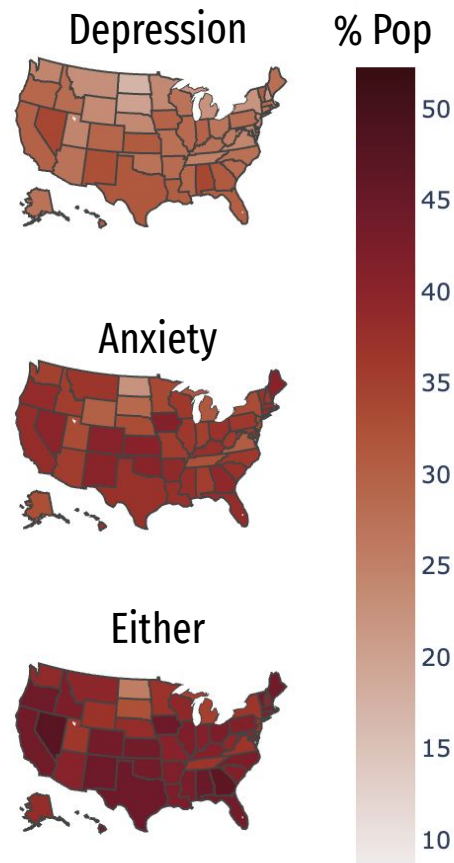




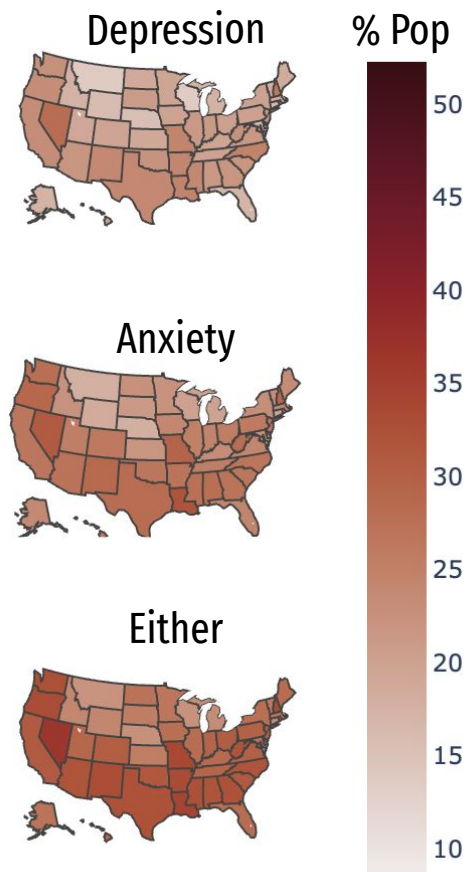
Peak-Trough-Peak



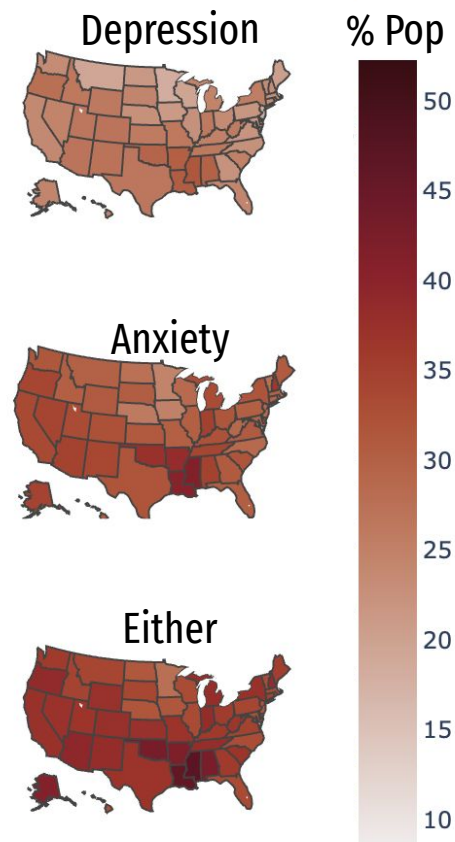
Nov 3, 2020



May 26, 2021

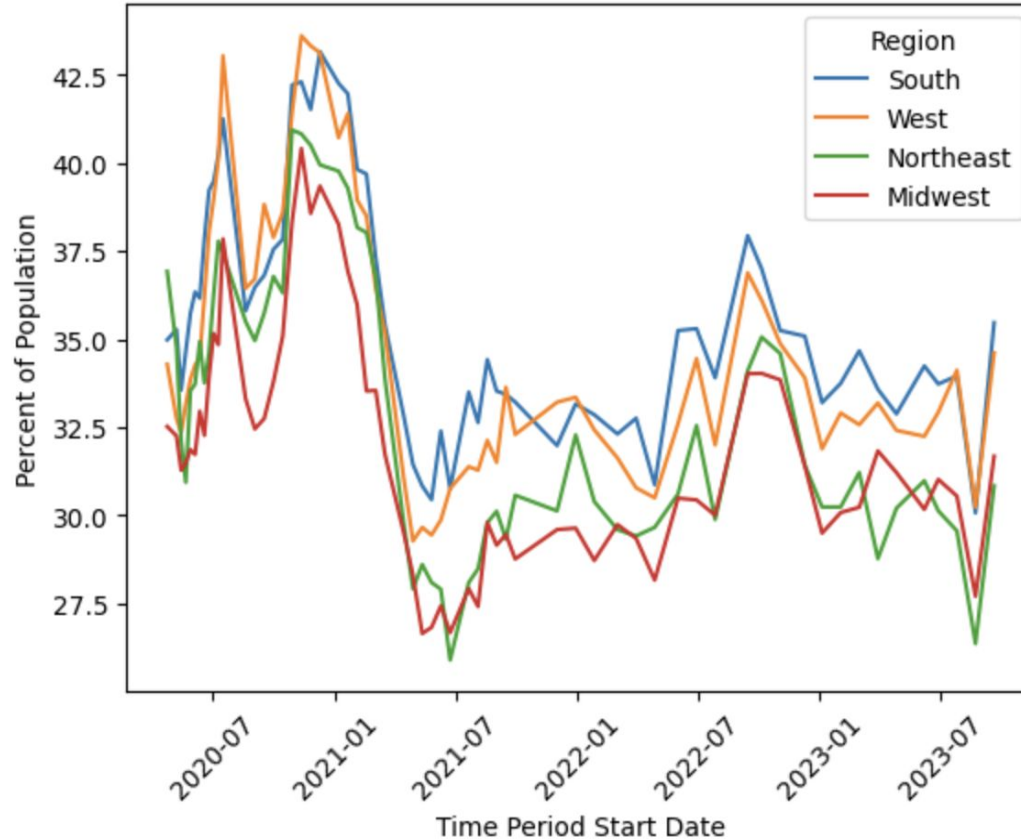


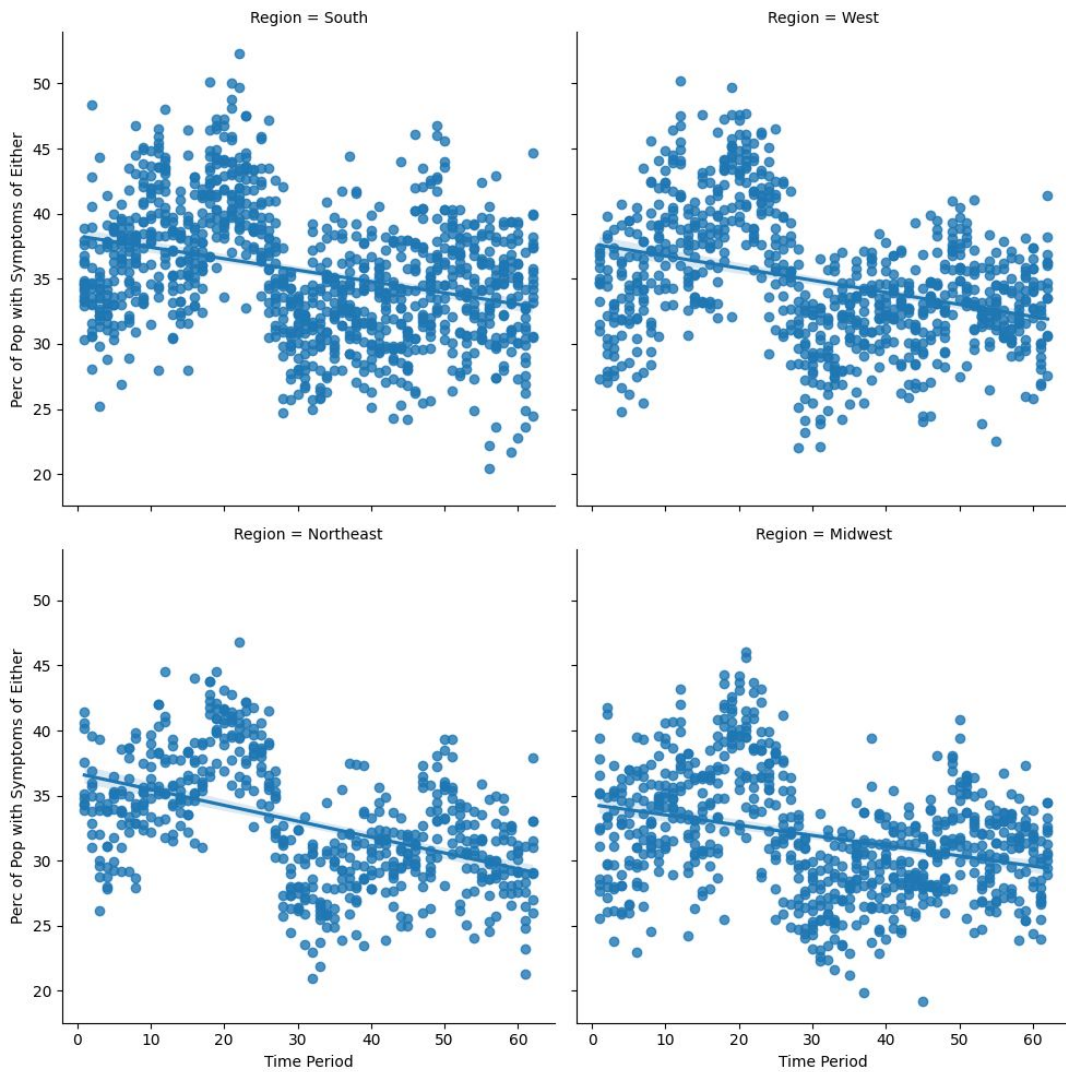
Sep 14, 2022



Regional Differences

Combined Symptoms Over Time by Region





Regression



Conclusion



Findings

- **Midwest and Northeast are consistently lower than the South and West.**
- Spike in anxiety and depression rates following the outbreak of COVID-19, followed by a decline to a steady rate.
- Overall Anxiety rates tended to be higher than depression



Potential Caveats

- Our regression model was not a great fit
 - Explained about 19% of variation
 - Data not linear
- Spikes in the data may be correlated with major world/US events, but there's no conclusive evidence to prove that they're causal

QUESTIONS?

